THE ARDENNES CAMPAIGN SIMULATION DATA BASE (ACSDB)

Final Report

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Sections I through II-I

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ABSTRACT

The Ardennes Campaign Simulation Data Base (ACSDB) is a computerized data base of the World War II 1944 campaign fought in December 1944 and January 1945, popularly known as the "Battle of the Bulge." The ACSDB uses the dBASE IV data base management software for storage and manipulation of the data. The data consists of daily information on unit location, activities, order of battle, personnel, equipment, and logistics at the army-, corps-, division-, and brigade-level for the period 16 December 1944 through 16 January 1945, plus data on tactical air operations, tables of organization and equipment (T/O&E) of units, and equipmemt used by opposing forces in the Ardennes Campaign. Data for the ACSDB was obtained from primary and secondary sources on file at libraries and archives in the United States, Great Britain, and the Federal Republic of Germany. The approximate size of the computer portion of the ACSDB is 39 megabytes. Additional material provided as part of the ACSDB includes a written draft final report with bibliographic, definitional, and other information; a set of full-scale color photographs of 1/100,000 scale maps of the Ardennes region with a clear acetate overlay on which are shown locations of US and German units on 16 December 1944; a user's guide for the ACSDB; a narrative on single-shot probability of kill (SSPK) data researched for the ACSDB; and a collection of photocopies of all records used for generation of the ACSDB totalling some ten cubic feet of printed material.

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ORGANIZATION OF THE ACSDB

The ACSDB is divided into eight data base files, each of which is designed to store and display all pertinent information on a particular aspect of the Ardennes battle. The eight data base files are:

Unit Data Base (File)
Unit Inventory Data Base (File)
Unit Location Data Base (File)
Air Data Base (File)
Table of Organization and Equipment (T/O&E) Data Base (File)
Weapons Data Base (File)
Reference Data Base (File)
Bibliography Data Base (File)

Attachment 2 illustrates the general organization of the ACSDB and the relationship of its components.

For purposes of discussion in the draft final report, the eight data base files are commonly referred to as "data bases," i.e, "Unit Data Base," "Unit Inventory Data Base," "Bibliography Data Base," etc. A data base (file) is composed of records, ranging in numbers from a minimum of 153 (Bibliography Data Base) to a maximum of 15,033 (Unit Inventory Data Base) records per data base. Each record is in turn composed of data fields, which contain identifying, statistical, or narrative information, such as a unit name, number of infantry personnel, or a description of a unit's operational mission.

The eight data bases in the ACSDB are:

Unit Data Base.

The Unit Data Base file contains personnel, medical, and logistical statistics of ground combat units of US, British, and German forces that participated in the Ardennes battle. Fields are also used in the Unit Data Base to record the order of battle of the US, British, and German units.

Unit Inventory Data Base.

The Inventory Data Base contains data on equipment (weapons and vehicles) strengths and losses of US, British, and German ground combat units. Abbreviations used for equipment designations in the Unit Inventory Data Base are identical to those used in the Table of Organization and Equipment (T/O&E) and Weapons Data Bases.

Unit Location Data Base.

In the Unit Location Data Base is recorded information on the location of US, British, and German ground combat units, including brigades, divisions, corps, and armies. The Unit Location Data Base is the narrative portion of the ACSDB and provides information on the geographical positions, activities, operations, missions, and commander status of ground combat units. Unit location is described utilizing Universal Transverse Mercator (UTM) grid coordinates used on the 1943 Geographical Section, General Staff maps (1/100,000 scale) produced by the Army Map Service. Full-scale color photographs of ten maps from this series, covering the Ardennes region, are provided to CAA as part of the deliverable.

Air Data Base.

The Air Data Base contains information on tactical air sorties flown by United States Army Air Forces, Royal Air Force, and Luftwaffe air units in the Ardennes battle area. Air Data Base information includes number of sorties, a description of operations and events during the missions, and geographical location of missions and airbases of origin of aircraft. Mission locations utilize the same map grid coordinates employed in the Unit Location Data Base.

Table of Organization and Equipment (T/O&E) Data Base.

Data from official Tables of Organization and Equipment (T/O&E) for all US, British, and German battalion-, regiment-, brigade-, and division-size units covered in the ACSDB are provided in the T/O&E Data Base, as well as some company-/troop-/battery-size units. Authorized personnel, equipment, and logistics strengths are provided for each unit in the T/O&E Data Base. The same equipment designations used in the Unit Inventory and Weapons Data Bases are used in the T/O&E Data Base.

Weapons Data Base.

In the Weapons Data Base is recorded information on equipment (vehicles and weapons) characteristics. Weapons Data Base records are used for all air and ground systems tracked in the Inventory and T/O&E Data Bases, including aircraft, armored fighting vehicles, indirect fire weapons, and softskin transport vehicles. Weapons characteristics include movement rates of vehicles and aircraft; caliber, rate of fire, and ammunition loads of artillery and tank guns; weights of systems; etc. The same equipment designations used in the Unit Inventory and T/O&E Data Bases are used in the Weapons Data Base.

Reference Data Base.

The Reference Data Base records the sources used for data and information in other data base files. Each record in the Unit Location, Air, T/O&E, and Weapons Data Bases contains a field identified as "Sources." An entry in this field matches identically an entry in the field "Source Number" in the Reference Data Base record which identifies the sources consulted for data and information in the data record.

Bibliography Data Base.

An entry under the field "Bibliography Name" in a Reference Data Base record in turn matches the entry under "Bibliography Name" in a Bibliography Data Base record. Full bibliographic citations for primary and secondary sources are provided in the Bibliography Data Base. Additional information in the written narratives of the draft final report supplement the Reference and Bibliography Date Bases for comprehensive referencing of all data recorded in the ACSDB.

ACSDB DATA RESEARCH AND COMPILATION

DMSi researchers examined primary and secondary data sources at archives and library facilities in the Washington, D.C. area and in Europe. DMSi's Fairfax, Virginia, office is conveniently located for research at the Washington area facilities which house many of the relevant records used for the compilation of data and information on the Ardennes battle. Foremost among these facilities is the National Archives annex in Suitland, Maryland -- the Washington Federal Records Center -- where operational records of US Army World War II units are stored. These include G-1, G-2, G-3, G-4, after-action, and historical file records of US armies, corps, and divisions, and separate field artillery, tank destroyer, and tank battalions. The majority of the data for the US forces that participated in the Ardennes campaign is derived from these records. The main building of the US National Archives in downtown Washington contains microfilm copies of German Army records collected at the end of World War II, as well as additional records of US Army units. The Office of the Center of Military History (OCMH), also located in downtown Washington, has an extensive collection of information on the Ardennes Campaign. Manuscripts on file at OCMH and prepared for Hugh M. Cole's official US Army history of the Ardennes were copied and used in DMSi's data compilation. Two Washington area libraries, the Library of Congress and the Pentagon Library, have extensive collections of unit histories of divisions participating in the Battle of the Bulge. These histories frequently prove to be valuable detailed secondary sources of information. Additional primary and secondary data sources are on file at the US Army Military History Institute Library at Carlisle Barracks, Carlisle, Pennsylvania. Finally, the Office of Air Force History at Bolling Air Force Base in Washington, D.C., was utilized for data on US tactical air operations in the Ardennes. DMSi researchers also located a valuable collection of microfilm records of selected US Army ground units at Bolling, including some not in the National Archives annex at Suitland.

Two DMSi researchers obtained data and information on British air and ground units in the Ardennes Campaign from primary sources at Her Majesty's Public Records Office in London. West German military archives in Freiburg and Koblenz were visited by a two-man DMSi research team to collect data on German Army units. Although the US Army acquired and shipped these records to the United States at the end of World War II, many of the operational records of German Army combat units were not microfilmed before the collection was returned to West Germany. DMSi researchers obtained relevant personnel, operations, and other records on units that took part in the Ardennes Campaign.

The mass of records assembled by DMSi researchers was catalogued and then analyzed to compile the detailed data required for the campaign-level data base. Of necessity, much data was interpolated, extrapolated, or otherwise derived from the various sources acquired from the research facilities. Records containing information pertinent to the ACSDB were sometimes kept on a weekly or monthly basis by the US Army during World War II. DMSi developed various "estimation methodologies" for generating daily data, using known data or experience to fill in gaps where data does not exist. The shortage of data and information for German units was more severe than that affecting US forces. Because of Germany's collapsing military situation at the end of 1944 and due to other factors, combat units either did not compile detailed reports or lost them in the defeat of the Germany Army. Some important monthly status reports (Zustandsbericht) for German armored and armored infantry divisions survived the war, as have other records, but by and large the German history of the Ardennes Campaign suffers from a lack of primary source records. This situation is somewhat ameliorated by the availability of postwar interviews with and manuscripts by German officers who participated in and planned the Ardennes offensive. Prepared by US Army historians, these reports provide an excellent overview of tactical operations by German units in the Ardennes, and fragmentary data on personnel and equipment strengths and attrition.

HISTORICAL ABSTRACT OF THE ARDENNES CAMPAIGN

This brief historical abstract is provided to give analysts a general history of the Ardennes Campaign. If more detailed information is required, several of the sources listed in a later section of this report should be consulted.

On 16 December 1944, three German armies (the Fifth Panzer (Armored), Sixth Panzer, and Seventh) launched a surprise offensive against a thinly-held section of the US front line in rugged, wooded terrain centered around the intersection of the borders of Luxembourg, Germany, and Belgium. This area, known as the Ardennes, was to become the focal point of a bitterly contested battle of several weeks' duration fought between units of the three German armies mentioned above and units of the US First and Third Armies and the British XXX Corps.

The Ardennes Campaign, popularly known as the "Battle of the Bulge," caught US units by almost complete surprise. By late 1944, the Allied High Command were confident that the Germans lacked the materiel and personnel resources to launch a major attack against Allied forces on the Western Front. However, under cover of strict security measures and a clever deception plan, the Germans managed to assemble an impressive force of armored, artillery, and infantry units without detection by the Allies. The overly ambitious German plan called for a penetration of the Allied lines in the same area used by the Germans in their victorious campaign in 1940, a swift advance to the port city of Antwerp which would split the British and US forces, and defeat the Allied armies in detail. Although impressive on paper, the forces assembled for the German attack suffered from a number of major weaknesses. Personnel quality was low, transportation networks were inadequate, and German air forces, although massed for the offensive, were wholly inadequate in face of the overwhelming air supremacy of the Allies. Nevertheless, aided by inclement weather conditions which grounded Allied aircraft, the Germans massed forces and penetrated the US line in several places, inflicting severe defeats on two US divisions in the opening phases of the battle.

After several days of major German penetrations, US forces rallied to slow and then stop the German attack. In fact, on the evening of the first day of the battle, US units on the northern shoulder of what was to become the "bulge" -- the salient created by the German attack -- had blunted critical attacks by some of the most formidable of the German panzer units, which had been ordered to reach the Meuse River line on the way to Antwerp by the end of the first day of the attack. Hard-fought defensive operations by US units at St. Vith and by the US 101st Airborne Division also hindered the German advance. From the south, by

beginning about 18 December, Lieutenant General George S. Patton's US Third Army attacked into the left shoulder of the bulge to relieve beleaguered US units at Bastogne. By Christmas Day, skies had cleared over the Ardennes region and the full might of Allied air supremacy was brought to bear on the German units in the bulge. By New Year's Day, the Germans had lost the initiative, the Allies counterattacked and approximately two weeks later, restored the front line in the Ardennes to its pre-16 December positions.

The German Ardennes Offensive was the last major offensive of the German Army on the Western Front. It cost the Germans heavily in personnel and equipment, irreplaceable losses which could not be made up from Germany's diminished manpower pool and crippled war industry. The attack was a desperate last gamble, whose chances for success from the start were minimal. Although US forces were badly defeated at the beginning of the attack, the eventual defeat of the German attack is attributable in large measure to the motivation of the individual American combat soldier, manifested from the very start of the attack.

The ACSDB contains data on operations from the period 16 December 1944-16 January 1945. Most historians agree that German failure was inevitable after late December, and 16 January is the approximate date on which the front line in the Ardennes were restored. German ground units included in the ACSDB belong to the German Fifth Panzer, Sixth Panzer, Seventh, and Fifteenth Armies, the latter army north of the main battle area and not directly involved in the campaign. US ground units included those of the First, Third, and Ninth Armies, the latter army opposite the German Fifteenth Army and only peripherally involved in combat operations in the Battle of the Bulge. British units are from the British XXX Corps.

Attachment 1

Units in the ACSDB

American --

2d Armored Division (AD)
3d AD
4th AD
5th AD
6th AD
7th AD
9th AD
10th AD
11th AD
17th Airborne Division (AbnD)
82d AbnD
101st AbnD
1st Infantry Division (ID)
2d ID
4th ID
5th ID
8th ID
9th ID
26th ID
28th ID
29th ID
30th ID
35th ID
75th ID
78th ID
80th ID
83d ID
84th ID
87th ID
90th ID
95th ID
99th ID
102d ID
104th ID
106th ID

British --

Guards Armoured Division
6th Airborne Division
43d Infantry Division (ID)
51st ID
53d ID
29th Armoured Brigade (ArmBde)

33d ArmBde
34th Tank Brigade

German --

1st SS Panzer Division (SSPzD)
2d SSPzD
9th SSPzD
10th SSPzD
12th SSPzD
27th SS Panzer Grenadier Division (SSPzGD)
28th SSPzGD
130th (Lehr) Panzer Division (PzD)
2d PzD
9th PzD
11th PzD
116th PzD
3d Panzer Grenadier Division (PzGD)
15th PzGD
3d Fallschirmjaeger Division (FJD)
5th FJD
150th Panzer Brigade
Fuehrer Begleit Brigade
Fuehrer Grenadier Brigade
9th Volks Grenadier Division (VGD)
12th VGD
18th VGD
26th VGD
47th VGD
62d VGD
79th VGD
167th VGD
212th VGD
246th VGD
272d VGD
276th VGD
277th VGD
326th VGD
340th VGD
344th VGD
352d VGD
363d VGD
560th VGD
59th Infantry Division (ID)
85th ID
89th ID
353d ID

In addition to the divisions, the ACSDB provides data for the separate combat units and service support units subordinated under corps and armies.

US armies and corps in the ACSDB are:

First Army
Third Army
Ninth Army
III Corps
V Corps
VII Corps
VIII Corps
XII Corps
XIII Corps
XVIII Airborne Corps
XIX Corps
XX Corps

The British corps in the ACSDB is:

XXX Corps

The German armies and corps include:

Fifth Panzer Army
Sixth Panzer Army
Seventh Army
Fifteenth Army
I SS Panzer Corps
II SS Panzer Corps
XII SS Corps
Corps Group Felber (renamed XIII Corps)
Corps Group Decker (provisionally organized)
XXXIX Panzer Corps
XLVII Panzer Corps
LVIII Panzer Corps
LIII Corps
LXVI Corps
LXVII Corps
LXXIV Corps
LXXX Corps
LXXXI Corps
LXXXV Corps

"Unit Name" (Field 1): Listed under "Unit Name" is the identification of the ground unit for which data is recorded in the Unit Data Base record. Data is recorded at three general echelons: division- (or independent armored brigade-) level, corps-level, and army-level.

Division-level data is subdivided into two parts, that pertaining to the entire division alone, and that describing any non-divisional combat unit(s) attached to the division. This format was utilized to facilitate the recording of data for units. Originally, it had been intended to incorporate the data of major detached organic divisional elements (such as Combat Command R of the 9th Armored Division detached from the 9th AD and attached to the 101st Airborne Division at Bastogne), as well as the data of attached non-divisional units, with parent divisions. This scheme proved to be entirely impractical for systematic recording of data and its proofing. Therefore, the data was split as described above. Entries such as "1st SSPzD", "Guards AD", and "106th ID" -- German 1st SS Panzer Division, British Guards Armoured Division, and US 106th Infantry Division -- under "Unit Name" indicate that the data recorded is for those divisions' organic components. Designations such as "1st SSPzD Att" or "28th ID Att" are used for any non-divisional combat unit(s) -- such as field artillery, tank, antitank, combat engineer, etc. -- attached to those divisions. Composition of divisional attachments is specified under the "ATTACHMENTS & DETACHMENTS" fields of the Unit Data Base which are described below.

Data for corps and armies is also subdivided into two parts, that describing non-divisional combat formations subordinated to corps or army -- field artillery, tank, antitank, combat engineer, etc. -- and that describing non-divisional service support units subordinated to corps or army -- the numerous quartermaster, medical, ordnance, and headquarters units which comprise the corps and army organizations of the German and Allied armies. Data is aggregated for these units, on a daily basis, in a single record. Entries under "Unit Name" describing non-divisional corps or army combat units include "Troops" as part of the unit name, for example "XLVII PzK Troops", "XXX Corps Troops", and "1st Army Troops" -- German XLVII Panzer Corps combat troops, British XXX Corps combat troops, and US First Army combat troops. Designations for the non-divisional corps or army service and support units include "HQ", in the case of US forces, or just the designation of the unit for the Germans and British. Examples are "3d Army HQ", "XVIII Corps HQ", "XXX Corps", "5th PzArmy", and "I SSPzK" -- US Third Army service support troops, US XVIII (Airborne) Corps service support troops, British XXX

Corps service support troops, German Fifth Panzer Army service support troops, and German I SS Panzer Corps (Korps) service support troops.

To assist in the identification of Allied and German units entered under "Unit Name" the following list of abbreviations is provided:

AD	armored division
PzD	panzer division
SSPzD	SS panzer division
PzGD	panzer grenadier division
SSPzGD	SS panzer grenadier division
ID	infantry division
VGD	volks grenadier division
AbnD	airborne division
FJD	fallschirmjaeger division
ArmBde	armoured brigade
Gds TkBde	Guards Tank Brigade
FBB	Fuehrer Begleit Brigade
FGB	Fuehrer Grenadier Brigade
TkBde	tank brigade
Corps	corps
K	corps
Korps Decker	German temporary administrative HQ formation
Korps Felber	German corps renamed XIII K (Corps) in January 1945
PzK	panzer corps
SSPzK	SS panzer corps
1st Army	US First Army
3d Army	US Third Army
9th Army	US Ninth Army
5th PzArmy	German Fifth Panzer Army
6th PzArmy	German Sixth (SS) Panzer Army
7th Army	German Seventh Army
15th Army	German Fifteenth Army
AG B	German Army Group B

"Date" (Field 2): Provided under "Date" is the date in month-day-year format. In general, for data recorded on German forces in the Unit Data Base, a day lasts from 0600 hours of the indicated date to 0600 hours of the following day. For US and British forces, the day is 0001 hours to 2400 hours of the

indicated date. See the ACSDB Time Conventions narrative for details.

"Nationality" (No field number): This is the nationality of the unit for which data is recorded. "A" designates American (US), "B" designates British, and "G" designates German.

"ON-HAND PERSONNEL --Total, --Armor, --Infantry, --Artillery, --Maintenance, --Maintenance-2, --Medical, --Transportation, --Supply, --Engineer, --Other" (Fields 3-12): A unit's total number of effective, or present for duty, personnel, is recorded under "ON-HAND PERSONNEL -- Total." The most equivalent German personnel strength category is "Tagesstaerke", or daily strength, which denotes all effective personnel with organic elements of a unit only.

The ten so-called military occupational specialty (MOS) categories (hereafter referred to as MOS categories for purposes of discussion) are used to record the number of effective personnel belonging to that MOS in a unit. MOS in the sense used in the ACSDB does not refer to specific military occupational specialty, except in the cases of critical maintenance MOSs, but rather to all personnel in a particular kind of unit. For example, all personnel in a separate US tank battalion, or in the self-propelled panzerjaeger (antitank) battalion of a German panzer division, are considered "Armor" personnel.

The definitions provided below are also used for the MOS categories in the ACSDB T/O&E Data Base. Note that in the Unit Data Base MOS strengths of divisions are provided only for US and German forces, and not for those of the British.

Terminology in the definitions includes the phrase "of regimental strength or below." This phrase pertains primarily to separate units. For organic divisional units, the phrase is not relevant. This means for example that the personnel of the component antitank company of the US infantry division's infantry regiment are not counted as artillery personnel, but rather as infantry. On the other hand, all personnel of a separate towed antitank company are counted as artillery personnel.

- Armor (or Combat Vehicle Crewmen in the T/O&E Data Base): All personnel in separate or organic divisional tank, armored, self-propelled tank destroyer, self-propelled antitank (panzerjaeger), cavalry, and armored reconnaissance units of regimental strength or below. Examples of units include the US cavalry group, the US self-propelled tank destroyer battalions, the tank battalion of the US armored division, the reconnaissance troop of the US infantry division, the self-propelled panzerjaeger abteilung (antitank battalion) of the German panzer division, the panzer aufklaerungs abteilung (armored reconnaissance battalion) of the German panzer division, the

self-propelled panzerjaeger kompanie (company) of the German volks grenadier division, and the German schwere panzer abteilung (heavy tank battalion). Personnel in the headquarters units of US tank destroyer and armored groups are also classified under the "Armor" MOS category.

- Infantry: All personnel in separate or organic divisional infantry, armored infantry, panzer grenadier, parachute infantry, glider infantry, ranger, and fallschirmjaeger (paratrooper) units of brigade strength or below. Examples of units include the US ranger battalion, the infantry regiment of the US infantry division, the fallschirmjaeger regiment of the German fallschirmjaeger division, and the panzer grenadier regiment of the German panzer grenadier division. Also categorized as infantry are personnel in miscellaneous German fortress units such as the fortress machine gun battalion and the fortress penal battalion (FstMGBn and FstPnBn), personnel in the feld ersatz bataillon (replacement battalion) of German divisions, and personnel in the US airborne division reconnaissance platoon.

- Artillery: All personnel in separate or organic divisional artillery, armored field artillery, towed tank destroyer or panzerjaeger, and chemical mortar units of regimental strength or below. Examples of units include US field artillery battalions, the divisional artillery of the US infantry division, US towed tank destroyer battalions, the panzerjaeger abteilung of the German volks grenadier division, and the US chemical mortar company. Also categorized as artillery are the personnel in the headquarters units of US field artillery groups and brigades.

- Maintenance: All personnel in separate or organic divisional ordnance maintenance battalions and companies. Examples include the US heavy automotive maintenance company, the ordnance company of the US airborne division, and the kraftfahrpark truppen (motor transport park troops) of German divisions.

- Maintenance-2: Maintenance-2 is a subset of Maintenance. Data for these personnel is provided only for US units in the ACSDB. These personnel are found in US divisional ordnance maintenance units and in US corps and army ordnance maintenance units. The following table shows current MOS descriptions and US Army World War II equivalents, the World War II equivalents used to calculate the Maintenance-2 personnel strengths in the ACSDB.

Current Description		World War II Equivalent	
Hawk Launcher Repairer	24L	--	
Vulcan Repairer	27F	--	
Chaparral/Vulcan Repairer	27G	--	
Lance Repairer	27L	--	
MLRS Repairer	27M	--	
Fire Control Instrument Repairer	41C	Fire Control Repairer	918
Metal Worker	44B	Metal Body Repairer	040
Machinist	44E	Machinist	114
Small Arms Repairer	45B	Small Arms Repairer	903
Fire Control System Repairer	45G	Fire Control Repairer	918
Tank Turret Repairer	45K	Turret Artillery Repairer	907
Artillery Repairer	45L	Artillery Repairer	913
Fuel & Electrical System Repairer	63G	Fuel & Electrical Repairer	912
Track Vehicle Repairer	63H	Track Vehicle Repairer	966

- Medical: All personnel in separate or organic divisional medical units of battalion strength or below. Examples of units are the sanitaets truppen (medical troops) of German divisions, the medical battalion of the US infantry division, the US field hospital, and the US collecting and clearing companies. Personnel of German veterinary units are also considered medical personnel.

- Transportation: All personnel in non-divisional tranportation units responsible for "long-haul" transport between divisions and higher echelons. These are primarily personnel in the US quartermaster truck company and its equivalent in the German and British armies. Although there exists a transportation component in US, British, and German divisions, it is not considered "long-haul" transport.

- Supply: All personnel in separate or organic divisional supply units. Examples of units include the quartermaster company of the US infantry division, the verwaltungs truppen (administrative and supply troops) of German divisions, and US quartermaster bakery and laundry companies.

- Engineer: All personnel in separate or organic divisional combat engineer, construction engineer, and pionier (engineer) units of regimental strength or below. Examples of units include the armored engineer regiment of the US armored division, the US combat engineer battalion, the US general service engineer regiment, and the pioniere (engineers) of German divisions. Personnel in the headquarters units of US engineer groups are also considered engineers.

- Other: All personnel not included in the MOS categories defined above. These include military police, divisional tranportation personnel, antiaircraft artillery personnel, bandsmen, and personnel in divisional headquarters and headquarters companies.

"REPLACEMENT PERSONNEL --Total, --Armor, --Infantry, etc." (Fields 13-22): Replacements are new personnel received by a unit during the day entered under "Date". For British units, all replacement and return to duty personnel are recorded under "REPLACEMENT PERSONNEL". Replacement personnel are recorded in total and by the MOS categories defined above.

"RETURNING PERSONNEL --Total, --Armor, --Infantry, etc." (Fields 23-32): Returning personnel are primarily personnel returning to duty from treatment at medical facilities above divisional echelon during the day entered under "Date". These may also include personnel returning from leave or detached duty. Personnel listed as "Returnees" or "Casuals" in US records are considered returning to duty personnel. Returning personnel are recorded in total and by the MOS categories defined above.

"CASUALTIES -- Total" (Field 33): Total casualties include the sum of the battle and non-battle casualties incurred by a unit during the day entered under "Date", i.e., killed, wounded, captured/MIA, and DNBI.

"Total Killed" (Field 34): These are the total number of personnel killed in action or died of wounds in a unit during the day entered under "Date".

"Total Wounded" (Field 35): These are the total number of personnel wounded or injured in action during the day entered under "Date". For US forces, lightly wounded in action, i.e., those personnel treated at divisional medical facilities not at higher echelons, are included. For German forces, the number includes only those personnel evacuated from and treated at

medical facilities above division, although this is not a confirmed precise definition of the term for German forces. For additional information on this field, see the various narratives which describe the compilation and estimation of US and German personnel data.

"Total Captured/MIA" (Field 36): These are the total number of personnel captured by the enemy or whose whereabouts following a combat action are unknown, whether through becoming separated from their unit, being wounded or killed or unidentified, or through desertion. These statistics are recorded for the day entered under "Date". For further information on the derivation of this data, see later discussions in this paper on US and German personnel data.

"Total DNBI" (Field 37): These are the total number of personnel who become sick or injured through causes other than combat on the day entered under "Date".

"FACILITIES & STATISTICS -- No. of Hospital Beds" (Field 38): This is the number of available fully supported hospital beds maintained at army level. This data is provided for US forces from statistics found in Office of the Surgeon General records, and is estimated for German forces using typical German hospital army-level organization. These statistics are recorded for the day entered under "Date".

"FACILITIES & STATISTICS -- No. of Beds Filled" (Field 39): This is the number of fully supported hospital beds maintained at army level, occupied by patients. This data is provided only for US forces, using statistics from the Surgeon General records. These statistics are recorded for the day entered under "Date".

"FACILITIES & STATISTICS -- No. of WIA Entering Hospital" (Field 40): This is the number of admissions of wounded in action personnel to hospitals maintained at army level. This data is provided only for US forces, using statistics from the Surgeon General records. These statistics are recorded for the day entered under "Date".

"FACILITIES & STATISTICS -- No. of DNBI Entering Hosptial" (Field 41): This is the number of admissions of DNBI (disease and non-battle injury cases) to hospitals maintained at army level. This data is provided only for US forces, using statistics from the Surgeon General records. These statistics are recorded for the day entered under "Date".

"FACILITIES & STATISTICS -- No. Died in Hospital" (Field 42): This is the number of hospital cases, in fully supported hospital beds maintained at army level, who died from all causes, i.e., wounds or DNBI. This data is provided only for US forces, using statistics from the Surgeon General records. These

statistics are recorded for the day entered under "Date".

"FACILITIES & STATISTICS -- No. Evacuated" (Field 59): This is the number of personnel evacuated from army-level hospitals for medical treatment at higher echelons. This data is provided only for US forces, using statistics from the Surgeon General records. These statistics are recorded for the day entered under "Date".

Unit Data Base Logistics Data (Fields 43-48, 54-55): Recorded in the logistics data section of the Unit Data Base is information on ammunition, other supply, and motor fuel or POL amounts stored, received, and expended or consumed by a unit on the day entered under "Date". For complete definitions of ammunition categories as they relate to the US, British, and German forces ("Tank/AT", "Artillery", "Special", and "Other Ammo"), consult the ACSDB T/O&E narrative.

Due to the variety of sources used for US, British, and German logistics data, the various methodologies employed for data estimation, and the different supply systems maintained by the Allied and German armies, it is necessary to refer to later sections of this paper for comprehensive descriptions of the supply data recorded in the ADCSB. The following are general definitions of terms.

"SUPPLY CAPACITY -- AMMO TYPE -- ON-HAND, REC'D, USED" (Fields 43-46): All ammunition amounts in the Unit Data Base are in hundreds of pounds. For divisions, ammunition on-hand ("AMMO TYPE -- ON-HAND") includes ammunition in divisional supply. For corps or armies, it is the ammunition stored in corps or army dumps and depots and does not include ammunition of subordinated divisions (or in the case of armies, subordinated corps or divisions). For divisions, ammunition received ("AMMO TYPE -- REC'D") includes ammunition supplied to or received by a division. For corps or armies, ammunition received is the amount received by a corps or army and does not include the ammunition received by subordinated divisions (or in the case of armies, subordinated corps or divisions). For divisions, ammunition used ("AMMO TYPE -- USED") includes ammunition expended or consumed by a division. For corps or armies, it is the ammunition expended or consumed by non-divisional corps or army units and the ammunition supplied to subordinated divisions (or in the case of armies, subordinated corps or divisions). As detailed in the T/O&E narrative, "Tank/AT" ammunition is primarily tank and antitank gun ammunition; "Arty" ammunition is ammunition of artillery guns and howitzers; "Special" ammunition is the ammunition of the German nebelwerfer rocket projectors; and "Other" ammunition is small arms, machine gun, antiaircraft artillery, and miscellaneous ammunition.

"Other Supply -- ON-HAND, REC'D, USED" (Field 47): For US units, these are the amounts of Class I, II, and IV supplies, as described in US Army Field Manual 101-10, 21 December 1944, page 308. The supply distribution scheme outlined above applies to "Other Supply" data. "Other Supply" data is provided in the Unit Data Base for US forces only.

"Fuel -- ON-HAND, REC'D, USED" (Field 48): These are the amounts of fuel, in thousands of gallons, stored, received, and expended or consumed by a unit. The supply distribution scheme outlined above applies to "Fuel" data. Fuel data for British units includes all petroleum, oil, and lubricants (POL). For Germans, the data pertains only to motor fuel. For US divisions, the data is for motor fuel only, and for US corps and armies it includes motor fuel and POL.

"Transportation Capacity (dry)/Transportation Capacity (wet)" (Fields 54 & 55): This is the total payload capacity in tons and thousands of gallons of transportation units responsible for long-haul transport between divisions and higher echelons. Details on the calculation of these capacities for US, British, and German forces are provided in later sections of this paper.

"ATTACHMENTS & DETACHMENTS -- TYPE, UNIT NAME" (Fields 60-79): Encoded in this section of the Unit Data Base is information critical to understanding the organization of and data contained in the Unit, Unit Inventory, and Unit Location Data Bases of the ACSDB. Fields 60-79 are used to record the daily order of battle of combat units in the US, British, and German armies during the period 16 December 1944-16 January 1945. Attached and detached units, divisional and non-divisional, are recorded under "ATTACHMENTS & DETACHMENTS", including company-/battery-/troop-size units and above. (Selected platoon-size units, such as US tank destroyer platoons, are also recorded under "ATTACHMENTS & DETACHMENTS".) It is important to note that a comprehensive listing of attachments and detachments of a unit on its first day of participation in the ACSDB is provided for that day. On subsequent days changes only in attachments and detachments of a unit are recorded. Two report features of the ACSDB (see description of the "ADLIST 1" and "ADLIST 2" programs in the ACSDB User's Guide under separate cover) allow analysts to view on computer screen or print out in hard copy the daily attachment and detachment changes of all units of a nationality (US, British, or German), or all attachment and detachment changes of a single unit by day. It is recommended that analysts print out hard copies of both of the ADLIST programs to utilize most fully the ACSDB.

Under "ATTACHMENTS & DETACHMENTS -- UNIT NAME" is identified the specific unit which is attached or detached. Examples of entries are "655th HyPjBN", "259th FA BN 4.5" G (T)", and "2/B/635th TD BN (T)" -- German 655th Heavy Panzerjaeger

Battalion, US 259th Field Artillery Battalion (towed 4.5" gun), US 2d Platoon, B Company, 635th Tank Destroyer Battalion (towed).

Attachments and detachments of are qualified in two ways: 1) by the absence of the term "(REF)" after the attachment or detachment name, or 2) by the inclusion of the term "(REF)" after the attachment or detachment name. The difference between the two qualifications is as follows.

- An entry recorded under "ATTACHMENTS & DETACHMENTS -- UNIT NAME" without the term "(REF)" after it means that the personnel, equipment, and logistics data for this unit in the Unit and Unit Inventory Data Base is recorded with the parent unit identified under "Unit Name." This qualification is used primarily with non-divisional corps- and army-level units such as tank, tank destroyer, and field artillery battalions.

- An entry recorded under "ATTACHMENTS & DETACHMENTS -- UNIT NAME" followed by the term "(REF)" means that the identified unit is an army, corps, division, or organic divisional unit, and that the personnel, equipment, and logistics data for this unit in the Unit and Unit Inventory Data Bases is <u>not</u> recorded with the parent unit identified under "Unit Name". Attachments and detachments identified with "(REF)" are marked as such for reference purposes only. In the case of armies/corps/divisions subordinated to army group, corps/divisions to armies, and divisions to corps, this information is provided primarily to describe order of battle organization. This is also the reason for identifying organic divisional attachments and detachments, although originally it had been intended to record the personnel, equipment, and logistics data of these units with the controlling division as is done with non-divisional units. Due to the impracticality of separating data of organic divisional units from data of the parent division, however, it was decided to qualify all attachments and detachments of organic divisional units with "(REF)".

For the Germans, attachments and detachments listed with armies and corps headquarters (I SSPzK, LIII K, 7th Army, etc.), include corps- and army-level service support units. The "Remarks" field (Field 57), should also be consulted for German armies and corps, as it was not possible to identify all service support units in Fields 60-79 due to space limitations. For US and British units, service support units are not identified in Fields 60-79, but instead are described in other narratives for the ACSDB. This is done because of the limited space available in the computer format, and the inability to list all of the many service support units in the limited format. The appropriate

narrative section of the report containing descriptive information on US service support organization is II-B ("US Non-Divisional Personnel Data"). Section II-F of the report ("British Personnel, Logistics, and Order of Battle Data Derivation") explains the derivation of the daily British order of battle. Due to time constraints and lack of information in sources, an abbreviated order of battle is provided for the British. Specific British service support elements are not identified in the "Attachments and Detachments" section of the ACSDB; only combat formations are included.

Entries under "ATTACHMENTS & DETACHMENTS -- TYPE" match the designations used for units in the T/O&E Data Base. Examples include "HyPjBN", "RcnReg", and "TDCoSP" -- German heavy panzerjaeger battalion, British reconnaissance regiment, and US self-propelled tank destroyer company. Some unit designations entered under "ATTACHMENTS & DETACHMENTS -- TYPE" are not in the T/O&E Data Base, because T/O&Es are not recorded for these units. These are principally units organic to divisions. An example is "PzAA", or panzer aufklaerungs abteilung -- German armored reconnaissance battalion. To assist in the identification of all designations recorded under "ATTACHMENTS & DETACHMENTS -- TYPE", Attachment 1 to this paper lists all designations with their full definitions.

Preceding all entries under "ATTACHMENT & DETACHMENTS -- TYPE" is either a "+" or a "-" sign. These signs indicate if the designated unit was attached (+) or detached (-) from the unit identified under "Unit Name" on the day entered under "Date".

Organization of the Allied and German armies as provided under the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base is the blueprint for understanding and interpreting the personnel, equipment, and logistics data recorded in the Unit and Unit Inventory Data Base. For example, if a corps controls five non-divisional field artillery battalions, the total tube strength recorded for the "corps troops" in the Unit Inventory Data Base will be 60 tubes, at 12 tubes per battalion and discounting any equipment attrition. The artillery personnel strength of the corps will reflect that of five artillery battalions, assuming no other artillery personnel such as chemical mortar and towed antitank troops are subordinated to the corps.

"Sources" (Field 56): Due to the considerable amount of estimation and the variety and number of sources necessary to generate data for the Unit Data Base, this paper serves as the sources reference for the Unit Data Base. To record in Reference Data Base format all bibliographic sources and estimation methodologies used for the Unit Data Base would have proved impractical and unwieldy. Later sections of this paper record in detail the sources and estimation processes employed to generate

data, by individual unit. Section XV of the report ("Comprehensive List of Data Sources Identified in Final Report") contains a list of all sources described in the written final report. Section XV and the Bibliography Data Base should be consulted if complete information on a data source is required.

"Remarks" (Field 57): Entered under "Remarks" is information in text format which qualifies and amplifies data in any other field in the record.

Attachment 1

Unit Abbreviations Used for ATTACHMENTS & DETACHMENTS
Section of Unit Data Base

US Units

AAABNG	Antiaircraft Artillery Battalion, Gun
AAABNAW	Antiaircraft Artillery Battalion, Automatic Weapons
AAABNSP	Antiaircraft Artillery Battalion, Self-Propelled
AAABtAW	Antiaircraft Artillery Battery, Automatic Weapons
AAABtG	Antiaircraft Artillery Battery, Gun
AAABtSP	Antiaircraft Artillery Battery, Self-Propelled
AAAGrpH	Antiaircraft Artillery Group, Headquarters
AbnD	Airborne Division
AD	Armored Division
ArmBde	Armored Brigade (Combat Command)
ArmFABN	Armored Field Artillery Battalion
ArmGrpH	Armored Group Headquarters
ArmInfB	Armored Infantry Battalion
ArmInfC	Armored Infantry Company
Army	Army
CavGrp	Cavalry Group
CavGrpH	Cavalry Group Headquarters
CavSq	Cavalry Squadron
CavTrp	Cavalry Troop
Corps	Corps
DIVARTY	Divisional Artillery
EngBN	Engineer Battalion
EngCo	Engineer Company
EngReg	Engineer Regiment
FABN105	105mm Howitzer Field Artillery Battalion (towed)
FABN155	155mm Gun or Howitzer Field Artillery Battalion (towed)
FABN240	240mm Howitzer Field Artillery Battalion (towed)
FABN4.5	4.5-inch Gun Field Artillery Battalion (towed)
FABN8"G	8-inch Gun Field Artillery Battalion (towed)
FABN8"H	8-inch Howitzer Field Artillery Battalion (towed)
FABN(P)	Field Artillery Battalion (provisional)
FABNSP	155mm Gun Field Artillery Battalion (self-propelled)
FABt155	155mm Gun or Howitzer Field Artillery Battery (towed or self-propelled)
FABtSP	155mm Gun Field Artillery Battery (self-propelled)
FAGrpHQ	Field Artillery Group Headquarters
FAObsCo	Field Artillery Observation Company
FRInfBN	French Infantry Battalion
FRInfCo	French Infantry Company
FRPIBN	French Parachute Infantry Battalion
GIBN	Glider Infantry Battalion
GIReg	Glider Infantry Regiment
ID	Infantry Division
InfBN	Infantry Battalion

InfBde	Infantry Brigade
InfReg	Infantry Regiment
MrtBN	Chemical Mortar Battalion
MrtCo	Chemical Mortar Company
PFABN	Parachute Field Artillery Battalion
PIBN	Parachute Infantry Battalion
PIReg	Parachute Infantry Regiment
RcnTrp	Reconnaissance Troop
RgrBN	Ranger Battalion
TDBNSP	Tank Destroyer Battalion (self-propelled)
TDBNT	Tank Destroyer Battalion (towed)
TDCoSP	Tank Destroyer Company (self-propelled)
TDCoT	Tank Destroyer Company (towed)
TDPltT	Tank Destroyer Platoon (towed)
TkBN	Tank Battalion
TkBNMX	Mine-Exploding Tank Battalion
TkCo	Tank Company
TkCoMX	Mine-Exploding Tank Company

British Units

AAABdeH	Antiaircraft Artillery Brigade Headquarters
AbnD	Airborne Division
ACReg	Armoured Car Regiment
AD	Armoured Division
AGRA	Army Group Royal Artillery
ALBde	Air Landing Brigade
AOPFlt	Air Observation Post Flight
APCReg	Armoured Personnel Carrier Regiment
ArmBde	Armoured Brigade
ArmReg	Armoured Regiment
ATBty	Antitank Battery
ATReg	Antitank Regiment
CrocReg	Crocodile (flamethrowing tank) Regiment
CrocSq	Crocodile (flamethrowing tank) Squadron
DIVARTY	Divisional Artillery
DIVRE	Divisional Royal Engineers
FldBty	Field Artillery Battery
FldCoRE	Field Company, Royal Engineers
FldReg	Field Artillery Regiment
FldSqRE	Field Squadron, Royal Engineers
Gds	Guards
ID	Infantry Division
InfBde	Infantry Brigade
InfBN	Infantry Battalion
HAAReg	Heavy Antiaircraft Artillery Regiment
HvyBty	Heavy Artillery Battery
LAAReg	Light Antiaircraft Artillery Regiment
MedReg	Medium Artillery Regiment
ParaBde	Parachute Brigade
RcnReg	Reconnaissance Regiment
SASSq	Special Air Service Squadron
TkBde	Tank Brigade
TkReg	Tank Regiment
TrkCo	Truck Company

German Units

AArtyBN	Army Artillery Battalion
AArtyBt	Army Artillery Battery
AG	Army Group
AMrtBN	Army Mortar Battalion
ArtBN	Artillery Battalion
ArtReg	Artillery Regiment
ASvcS	Army Service School
BN	Battalion
BrB(GE)	Light bridge equipment column (company) non-motorized
BrB(m)	Light bridge column (company) fully motorized
BrJ(GE)	Heavy bridge equipment column (company) non-motorized
BrJ(m)	Heavy bridge column (company) fully motorized
Bty	Battery
Co	Company
EngBde	Engineer Brigade
EngBN	Combat Engineer Battalion
EngBN C	Construction Engineer Battalion
EngBrBN	Engineer Bridge Battalion
EngCo	Engineer Company
EngEABN	Engineer Ersatz und Ausbildung (Replacement and Training) Battalion
EngRegC	Construction Engineer Regiment
FBB	Fuehrer Begleit (Escort) Brigade
FEB	Feld Ersatz Abteilung (Field Replacement Battalion)
FGB	Fuehrer Grenadier Brigade
FJArtR	Fallschirmjaeger (Paratrooper) Artillery Regiment
FJBN	Fallschirmjaeger (Paratrooper) Battalion
FJD	Fallschirmjaeger (Paratrooper) Division
FJEngBN	Fallschirmjaeger (Paratrooper) Engineer Battalion
FJMtrBN	Fallschirmjaeger (Paratrooper) Mortar Battalion
FJPjBN	Fallschirmjaeger Panzerjaeger (Paratrooper Antitank) Battalion
FJReg	Fallschirmjaeger (Paratrooper) Regiment
FJStBde	Fallschirmjaeger Sturmgeschuetz (Paratrooper Assault Gun) Brigade
FLAKBde	Fliegerabwehrkanone (Antiaircraft) Brigade
FLAKBN	Fliegerabwehrkanone (Antiaircraft) Battalion
FLAKCo	Fliegerabwehrkanone (Antiaircraft) Company
FLAKD	Fliegerabwehrkanone (Antiaircraft) Division
FLAKReg	Fliegerabwehrkanone (Antiaircraft) Regiment
FLAKStB	Fliegerabwehrkanone Sturm (Antiaircraft Assault) Battalion
FLAKStR	Fliegerabwehrkanone Sturm (Antiaircraft Assault) Regiment
FstArBN	Festung (Fortress) Artillery Battalion

FstArBt	Festung (Fortress) Artillery Battery
FstBN	Festung (Fortress) Battalion
FstMGBN	Festung (Fortress) Machine Gun Battalion
FstPAKC	Festung Panzerabwehrkanone (Fortress Antitank) Company
FstPKBN	Festung Panzerabwehrkanone (Fortress Antitank) Battalion
FstPnBN	Festung (Fortress) Penal Battalion
FusBN	Fusilier (Rifle) Battaliom
FusCo	Fusilier (Rifle) Company
FusR	Fusilier (Rifle) Regiment
GrenBN	Grenadier (Infantry) Battalion
Grp	Group
HG	Hermann Goering (used in reference to German fallschirmjaeger (paratrooper) units)
HyPjBN	Heavy Panzerjaeger (Antitank) Battalion
HyPzBN	Heavy Panzer (Armored) Battalion
ID	Infantry Division
InfBN	Infantry Battalion
InfReg	Infantry Regiment
K	Korps (Corps)
KG	Kampfgruppe (Battle Group)
OKW	Oberkommando der Wehrmacht (High Command of Armed Forces)
OTBde	Organization Todt (paramilitary civilian construction corps) Brigade
OTReg	Organization Todt (paramilitary civilian construction corps) Regiment
PjBN	Panzerjaeger (Antitank) Battalion
PjBNT	Towed Panzerjaeger (Antitank) Battalion
PjCo	Panzerjaeger (Antitank) Company
PzAA	Panzer Aufklaerungsabteilung (Armored Reconnaissance Battalion)
PzAR	Panzer (Armored) Artillery Regiment
PzArmy	Panzer (Armored) Army
PzBde	Panzer (Armored) Brigade
PzBN	Panzer (Armored) Battalion
PzCo	Panzer (Armored) Company
PzD	Panzer (Armored) Division
PzEngCo	Panzer (Armored) Engineer Company
PzGD	Panzer Grenadier (Armored Infantry) Division
PzGR	Panzer Grenadier (Armored Infantry) Regiment
PzK	Panzer Korps (Armored Corps)
PzLehrD	Panzer Lehr (Armored "Instruction") Division
PzReg	Panzer (Armored) Regiment
RRArtBN	Railway Artillery Battalion
RRArtBt	Railway Artillery Battery
SS	Schutzstaffeln (indicates combat elements of SS -- Waffen SS)
SSArtBN	SS Artillery Battalion
SSFLABN	SS Fliegerabwehrkanone (Antiaircraft) Battalion
SSFLABt	SS Fliegerabwehrkanone (Antiaircraft) Battery

SSHPzBN	SS Heavy Panzer (Armored) Battalion
SSK	SS Korps (Corps)
SSNwBN	SS Nebelwerfer (Rocket Launcher) Battalion
SSPzAA	SS Panzer Aufklaerungsabteilung (SS Armored Reconnaissance Battalion)
SSPzAR	SS Panzer (Armored) Artillery Regiment
SSPzBN	SS Panzer (Armored) Battalion
SSPzD	SS Panzer (Armored) Division
SSPzGD	SS Panzer Grenadier (Armored Infantry) Division
SSPzGR	SS Panzer Grenadier (Armored Infantry) Regiment
SSPzK	SS Panzer Korps (Armored Corps)
SSPzR	SS Panzer (Armored) Regiment
SSWfrBN	SS Werfer (Rocket Launcher) Battalion
StgBde	Sturmgeschuetz (Assault Gun) Brigade
StgCo	Sturmgeschuetz (Assault Gun) Company
StMrsCo	Sturm Moerser (Assault Mortar) Company
StPzBN	Sturm Panzer (Assault Vehicle) Battalion
VAK	Volksartillerie Korps ("People's" Artillery Corps)
VABN	Volksartillerie ("People's" Artillery) Battalion
VGBN	Volks Grenadier ("People's" Infantry) Battalion
VGD	Volks Grenadier ("People's" Infantry) Division
VGR	Volks Grenadier ("People's" Infantry) Regiment
VTB	Volga Tartar Battalion
VWB	Volkswerfer ("People's" Rocket Launcher) Brigade
VWBN	Volkswerfer ("People's" Rocket Launcher) Battalion
VWBty	Volkswerfer ("People's" Rocket Launcher) Battery
VWR	Volkswerfer ("People's" Rocket Launcher) Regiment